

Session 1.4: Scale Dynamics & Winner-Takes-All

Market tipping, HHI concentration analysis, switching frictions, and Uber China vs. Didi.

PART 1 (LECTURE)
60 Minutes

PART 2 (LAB PRACTICE)
60 Minutes

PART 3 (EXECUTIVE MEMO)
60 Minutes

STANDARD
Audited SEC Filings

Learning Objectives & Executive Overview

Clear conceptual grounding for undergraduate and graduate business students.

CORE COMPETENCIES

Goals

What You Will Master Today

- **Theoretical Mastery:** Evaluate seminal literature and analytical frameworks governing platform competition.
- **Empirical Rigor:** Reconcile management strategic claims with legally audited statutory filings.
- **Data Modeling:** Execute spreadsheet calculations with explicit coordinates and formula logic.
- **Executive Formulation:** Deliver structured board-level recommendations resolving complex trade-offs.

WINNER-TAKES-ALL DYNAMICS

Vocabulary

Key Topical Foundations

- **Arthur (1989)::** Path dependency, lock-in, and historical tipping.
- **Eisenmann Criteria::** High multi-homing costs & strong cross-side effects.
- **HHI Metric::** Herfindahl-Hirschman Index industry concentration.
- **Micro-Examples::** Search engines (monopoly), mobile OS, ride apps.

Theoretical Foundations: Winner-Takes-All Dynamics

Rigorous analytical frameworks and literature benchmarks.

THEORETICAL ARCHITECTURE

Literature

Seminal Framework Insights

- **Arthur (1989)::** Path dependency, lock-in, and historical tipping.
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STRATEGIC APPLICATION

Analysis

Business Model Mechanics

- **Billion-Dollar War::** Subsidy battle across China (2014-2016).
- **Homogeneous Demand::** Riders & drivers switching for pennies.
- **Failure to Tip::** Scale alone failed to produce monopoly pricing power.
- **Strategic Truce::** Uber sold China ops to Didi for 18.8% equity.

STRATEGIC TAKEAWAY

Market tipping, HHI concentration analysis, switching frictions, and Uber China vs. Didi.

Enterprise Case Dissection: Uber China vs. Didi

Historical case evidence and multi-year corporate trajectory analysis.

CORPORATE DYNAMICS

Timeline

Market Conditions & Strategic Moves

- *Billion-Dollar War::* Subsidy battle across China (2014-2016).
- *Homogeneous Demand::* Riders & drivers switching for pennies.
- *Failure to Tip::* Scale alone failed to produce monopoly pricing power.
- *Strategic Truce::* Uber sold China ops to Didi for 18.8% equity.

ANALYTICAL FOCUS

Evidence

Operational & Financial Findings

- *Market Share Data::* Search, Mobile OS, and US Ride-Hailing shares.
- *HHI Calculation::* Sum of squared market shares (SUM(Share^2)).
- *Concentration Levels::* Unconcentrated (<1500) vs. High (>2500).
- *Result::* Search HHI > 7300, Mobile OS > 5800, Ride-Hailing ~6150.

Lab Dataset & Empirical Metrics: Applied Excel Lab

Reconciling corporate filings with explicit spreadsheet formulas.

SPREADSHEET MODELING

Lab Spec

Data Metrics & Formulas

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MANAGEMENT DILEMMA

Decision Lab

Strategic Tension

- **Margin Divergence::** Why Google makes 30%+ margin while Uber struggled.
- **Multi-Homing Trap::** High concentration with zero switching barriers.
- **Defensible Moats::** Three design choices creating genuine user lock-in.
- **Executive Memo::** Assessing platform pricing power beyond market share.

Executive Problem Solving: Executive Lab

Framing complex management dilemmas and evaluating corporate trade-offs.

EXECUTIVE CHALLENGE

Dilemma

Corporate Strategic Tension

- *Margin Divergence::* Why Google makes 30%+ margin while Uber struggled.
- *Multi-Homing Trap::* High concentration with zero switching barriers.
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EVALUATION RUBRIC

Requirements

Board-Level Deliverable

- *Actionable Recommendation:* Formulate a concrete 3-bullet decision brief for senior leadership.
- *Quantitative Defense:* Support recommendations using audited financial metrics from the lab spreadsheet.
- *Risk Mitigation:* Address channel friction, partner resistance, or competitive retaliation.
- *Implementation Timeline:* Define immediate 90-day execution milestones.

Lab Transition: 120-Minute Hands-On Execution

Transitioning from lecture observation to hands-on data manipulation and strategic decision-making.

EXERCISE 1: FOUNDATIONAL PRACTICE

60 Min Lab

Applied Excel Lab

Hands-on spreadsheet protocol using audited historical numbers:

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- **HHI Calculation::** Sum of squared market shares ($\text{SUM}(\text{Share}^2)$).
- **Concentration Levels::** Unconcentrated (<1500) vs. High (>2500).
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EXERCISE 2+: ADVANCED PROBLEM SOLVING

60 Min Memo

Executive Lab

Applied scenario analysis, trade-off evaluation, and executive brief:

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- **Multi-Homing Trap::** High concentration with zero switching barriers.
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- **Executive Memo::** Assessing platform pricing power beyond market share.